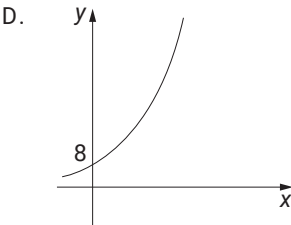
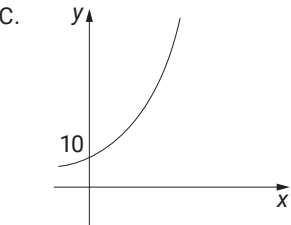
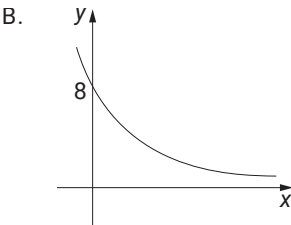
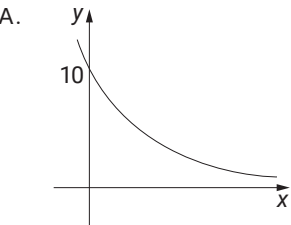


- A.** 4  
**B.** 8  
**C.** 32  
**D.** 64

3. Functions, 2ADV F2 2021 HSC 5 MC

Which of the following best represents the graph of  $y = 10(0.8)^x$ ?



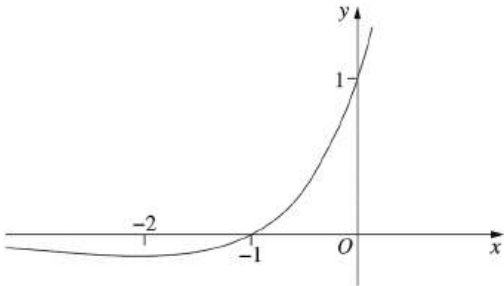
4. L&E, 2ADV E1 2021 HSC 3 MC

Which of the following represents the domain of the function  $f(x) = \ln(1 - x)$ ?

- A.  $[1, \infty)$
- B.  $(1, \infty)$
- C.  $(-\infty, 1]$
- D.  $(-\infty, 1)$

5. L&E, 2ADV E1 2015 HSC 8 MC

The diagram shows the graph of  $y = e^x(1 + x)$ .



How many solutions are there to the equation  $e^x(1 + x) = 1 - x^2$ ?

- A. 0
- B. 1
- C. 2
- D. 3

6. L&E, 2ADV E1 SM-Bank 2

The population of Indian Myna birds in a suburb can be described by the exponential function

$$N = 35e^{0.07t}$$

where  $t$  is the time in months.

- i. What will be the population after 2 years? (1 mark)
- ii. Draw a graph of the population. (2 marks)

7. Functions, 2ADV F2 SM-Bank 1

- i. Draw the graph  $y = \ln x$ . (1 mark)
- ii. Explain how the above graph can be transformed to produce the graph

$$y = 3\ln(x + 2)$$

and sketch the graph, clearly identifying all intercepts. (3 marks)

8. L&E, 2ADV E1 SM-Bank 14

The spread of a highly contagious virus can be modelled by the function

$$f(x) = \frac{8000}{1 + 1000e^{-0.12x}}$$

Where  $x$  is the number of days after the first case of sickness due to the virus is diagnosed and  $f(x)$  is the total number of people who are infected by the virus in the first  $x$  days.

- i. Calculate  $f(0)$ . (1 mark)
- ii. Find the value of  $f(365)$  and interpret it result. (2 marks)

9. Calculus, 2ADV C3 2022 HSC 27

Let  $f(x) = xe^{-2x}$ .

It is given that  $f'(x) = e^{-2x} - 2xe^{-2x}$ .

- a. Show that  $f''(x) = 4(x - 1)e^{-2x}$ . (2 marks)
- b. Find any stationary points of  $f(x)$  and determine their nature. (2 marks)
- c. Sketch the curve  $y = xe^{-2x}$ , showing any stationary points, points of inflection and intercepts with the axes. (3 marks)

10. Functions, 2ADV F2 SM-Bank 13

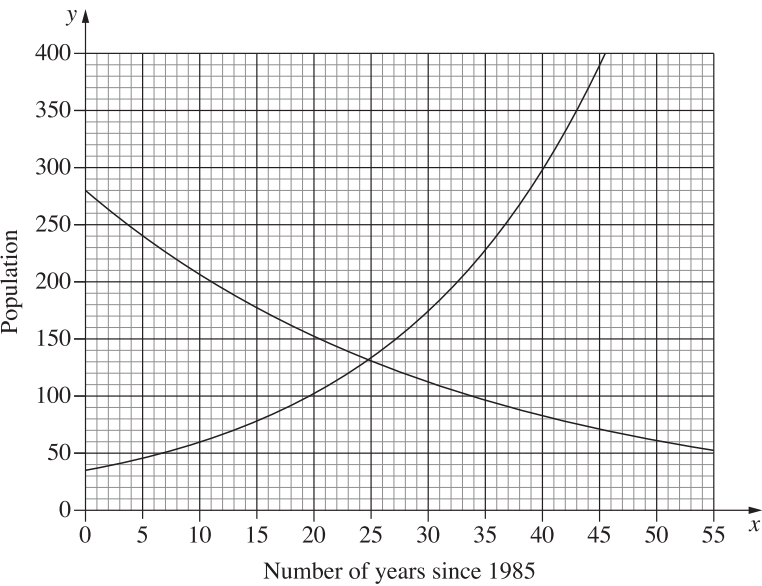
- i. Show that the function  $y = \frac{1 - e^x}{1 + e^x}$  is an odd function? (1 mark)
- ii. Sketch  $y = \frac{1 - e^x}{1 + e^x}$ , labelling all intercepts and asymptotes. (2 marks)

11. L&E, 2ADV E1 2024 HSC 13

The graph shows the populations of two different animals,  $W$  and  $K$ , in a conservation park over time. The  $y$ -axis is the size of the population and the  $x$ -axis is the number of years since 1985 .

Population  $W$  is modelled by the equation  $y = A \times (1.055)^x$ .

Population  $K$  is modelled by the equation  $y = B \times (0.97)^x$ .



Complete the table using the information provided. (3 marks)

|                                            | Population $W$ | Population $K$ |
|--------------------------------------------|----------------|----------------|
| Population in 1985                         | $A = 34$       | $B =$          |
| Percentage yearly change in the population |                |                |
| Predicted population when $x = 50$         |                | 61             |

12. Algebra, STD2 A4 2004 HSC 26a

- i. The number of bacteria in a culture grows from 100 to 114 in one hour.  
What is the percentage increase in the number of bacteria? **(1 mark)**
- ii. The bacteria continue to grow according to the formula  $n = 100(1.14)^t$ , where  $n$  is the number of bacteria after  $t$  hours.  
What is the number of bacteria after 15 hours? **(1 mark)**

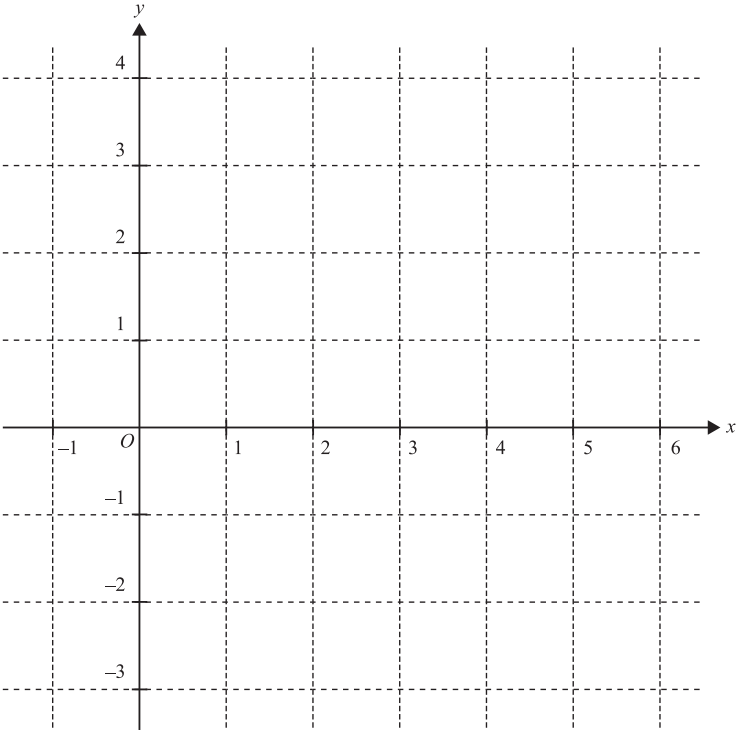
|                            |     |     |     |    |
|----------------------------|-----|-----|-----|----|
| Time in hours ( $t$ )      | 0   | 5   | 10  | 15 |
| Number of bacteria ( $n$ ) | 100 | 193 | 371 | ?  |

- iii. Use the values of  $n$  from  $t = 0$  to  $t = 15$  to draw a graph of  $n = 100(1.14)^t$ .  
Use about half a page for your graph and mark a scale on each axis. **(4 marks)**
- iv. Using your graph or otherwise, estimate the time in hours for the number of bacteria to reach 300. **(1 mark)**
- 

13. Calculus, 2ADV C3 2019 MET1 4

Given the function  $f(x) = \log_e(x - 3) + 2$ ,

- a. State the domain and range of  $f(x)$ . **(1 mark)**
- b. i. Find the equation of the tangent to the graph of  $f(x)$  at  $(4, 2)$ . **(2 marks)**
- ii. On the axes below, sketch the graph of the function  $f(x)$ , labelling any asymptote with its equation.  
Also draw the tangent to the graph of  $f(x)$  at  $(4, 2)$ . **(4 marks)**



14. Calculus, 2ADV C3 2023 HSC 30

Let  $f(x) = e^{-x}\sin x$ .

- a. Find the coordinates of the stationary points of  $f(x)$  for  $0 \leq x \leq 2\pi$ . You do NOT need to check the nature of the stationary points. (3 marks)
- b. Without using any further calculus, sketch the graph of  $y = f(x)$ , for  $0 \leq x \leq 2\pi$ , showing stationary points and intercepts. (2 marks)



15. Algebra, STD2 A4 2012 HSC 30c

In 2010, the city of Thagoras modelled the predicted population of the city using the equation

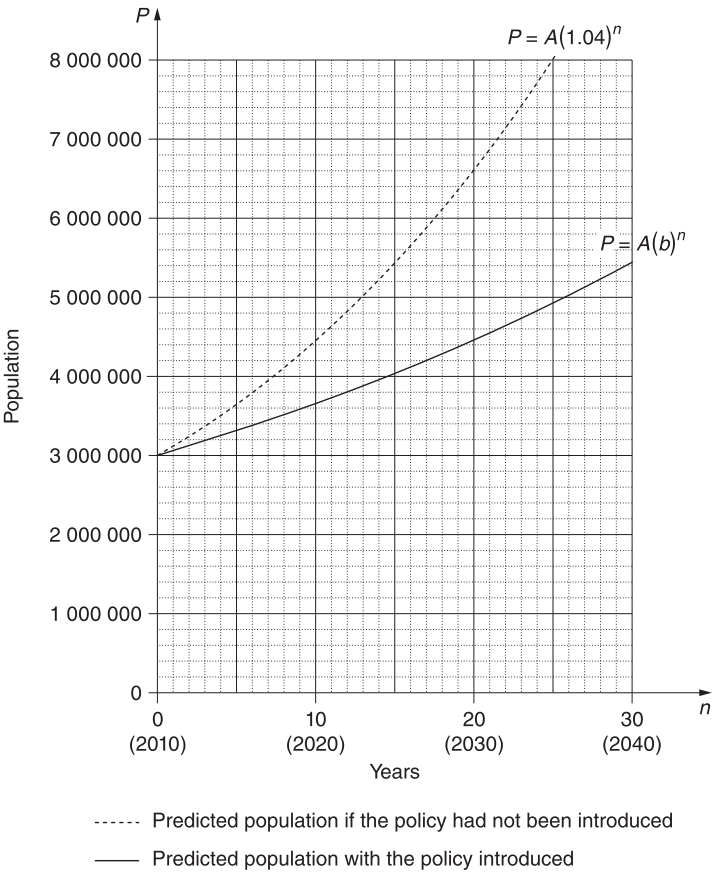
$$P = A(1.04)^n.$$

That year, the city introduced a policy to slow its population growth. The new predicted population was modelled using the equation

$$P = A(b)^n.$$

In both equations,  $P$  is the predicted population and  $n$  is the number of years after 2010.

The graph shows the two predicted populations.



- i. Use the graph to find the predicted population of Thagoras in 2030 if the population policy had NOT been introduced. (1 mark)
- ii. In each of the two equations given, the value of  $A$  is 3 000 000.  
What does  $A$  represent? (1 mark)
- iii. The guess-and-check method is to be used to find the value of  $b$ , in  $P = A(b)^n$ .
  - (1) Explain, with or without calculations, why 1.05 is not a suitable first estimate for  $b$ . (1 mark)
  - (2) With  $n = 20$  and  $P = 4\,460\,000$ , use the guess-and-check method and the equation  $P = A(b)^n$  to estimate the value of  $b$  to two decimal places. Show at least TWO estimate values for  $b$ , including

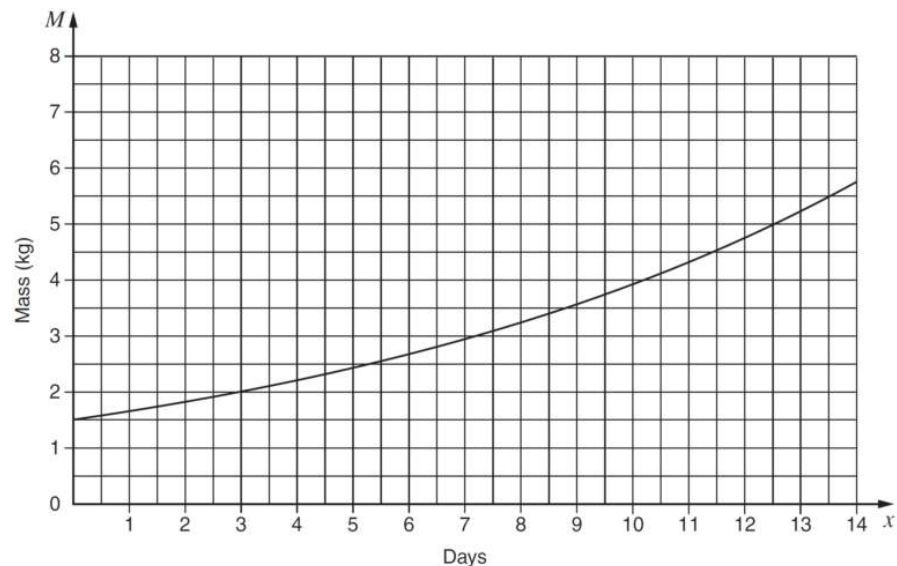
calculations and conclusions. (2 marks)

- iv. The city of Thagoras was aiming to have a population under 7 000 000 in 2050. Does the model indicate that the city will achieve this aim?

Justify your answer with suitable calculations. (2 marks)

## 16. Algebra, STD2 A4 2016 HSC 29b

The mass  $M$  kg of a baby pig at age  $x$  days is given by  $M = A(1.1)^x$  where  $A$  is a constant. The graph of this equation is shown.



- i. What is the value of  $A$ ? (1 mark)
- ii. What is the daily growth rate of the pig's mass? Write your answer as a percentage. (1 mark)

## Worked Solutions

### 1. L&E, 2ADV' E1 2019 HSC 1 MC

$$4 - x > 0$$

$$-x > -4$$

$$x < 4$$

$$\Rightarrow A$$

### 2. Algebra, STD2 A4 SM-Bank 1 MC

From graph:

$$a^5 = 4$$

$$\therefore 2(a^5) = 2 \times 4$$

$$= 8$$

$$\Rightarrow B$$

### 3. Functions, 2ADV F2 2021 HSC 5 MC

By elimination:

$$\text{When } x = 0, y = 10(0.8)^0 = 10$$

→ Eliminate B and D

$$\text{As } x \rightarrow \infty, y \rightarrow 0$$

→ Eliminate C

$$\Rightarrow A$$

### 4. L&E, 2ADV E1 2021 HSC 3 MC

$$1 - x > 0$$

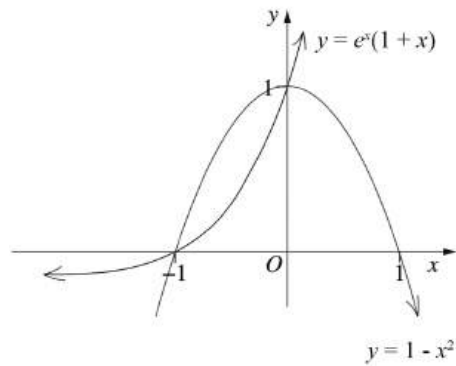
$$x < 1$$

$$x \in (-\infty, 1)$$

$$\Rightarrow D$$

Mean mark 52%!

5. L&E, 2ADV E1 2015 HSC 8 MC

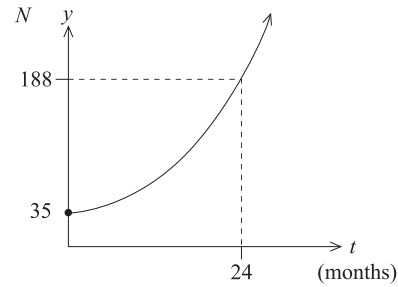


The graphs intersect at 2 points  
 $\therefore e^x(1+x) = 1-x^2$  has 2 solutions  
 $\Rightarrow C$

6. L&E, 2ADV E1 SM-Bank 2

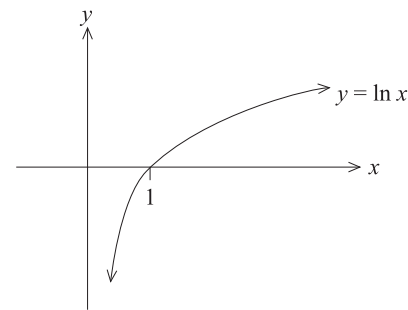
- i.  $N = 35e^{0.07t}$   
Find  $N$  when  $t = 24$ :  
 $N = 35e^{0.07 \times 24}$   
 $= 35e^{1.68}$   
 $= 187.79 \dots$   
 $= 188$  birds

ii.



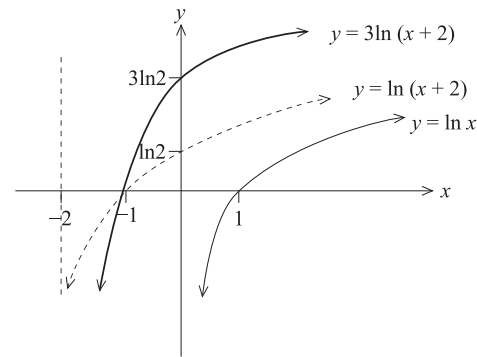
7. Functions, 2ADV F2 SM-Bank 1

i.



- ii. Transforming  $y = \ln x \Rightarrow y = \ln(x+2)$   
 $y = \ln x \Rightarrow$  shift 2 units to left.

Transforming  $y = \ln(x+2)$  to  $y = 3\ln(x+2)$   
 $\Rightarrow$  increase each  $y$  value by a factor of 3



8. L&E, 2ADV E1 SM-Bank 14

i.  $f(0) = \frac{8000}{1 + 1000e^0}$   
 $= \frac{8000}{1001}$   
 $= 7.99 \dots$

ii.  $f(365) = \frac{8000}{1 + 1000e^{-0.12 \times 365}}$   
 $= \frac{8000}{1 + 1000e^{-43.8}}$   
 $\approx 8000$

After 1 year, the model predicts the total number of people infected by the virus is 8000.

9. Calculus, 2ADV C3 2022 HSC 27

a.  $f'(x) = e^{-2x} - 2xe^{-2x}$   
 $f''(x) = -2e^{-2x} - [2x(-2)e^{-2x} + 2e^{-2x}]$   
 $= -2e^{-2x} + 4xe^{-2x} - 2e^{-2x}$   
 $= 4xe^{-2x} - 4e^{-2x}$   
 $= 4(x - 1)e^{-2x} \dots \text{as required}$

b. S.P.'s occur when  $f'(x) = 0$

$e^{-2x} - 2xe^{-2x} = 0$   
 $e^{-2x}(1 - 2x) = 0$

$e^{-2x} = 0 \Rightarrow \text{No solution}$   
 $1 - 2x = 0 \Rightarrow x = \frac{1}{2}$   
 $f''\left(\frac{1}{2}\right) = 4\left(\frac{1}{2} - 1\right)e^{-2 \times \frac{1}{2}}$   
 $= -2e^{-1} < 0 \Rightarrow \text{MAX}$

$f\left(\frac{1}{2}\right) = \frac{1}{2}e^{-1} = \frac{1}{2e}$

c. POI occurs when  $f''(x) = 0$

$f''(x) = 4(x - 1)e^{-2x} = 0 \Rightarrow x = 1$

Test for change in concavity:

$f''(0) = 4(0 - 1)e^0 = -4$

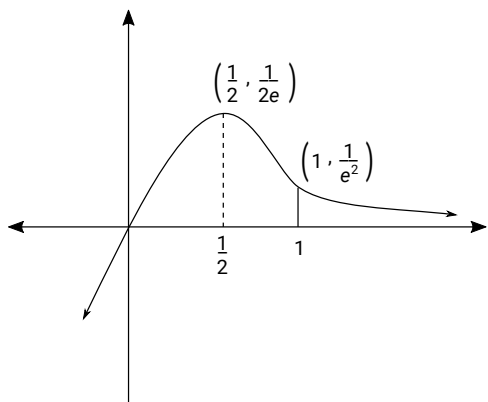
$f''(2) = 4(2 - 1)e^{-4} > 0 \Rightarrow \text{Concavity changes}$

$\therefore$  POI exists at  $\left(1, \frac{1}{e^2}\right)$

As  $x \rightarrow \infty \Rightarrow y \rightarrow 0^+$

Mean mark part (c) 54%.





## 10. Functions, 2ADV F2 SM-Bank 13

i.  $f(x) = \frac{1-e^x}{1+e^x}$

$$\begin{aligned} f(-x) &= \frac{1-e^{-x}}{1+e^{-x}} \times \frac{e^x}{e^x} \\ &= \frac{e^x-1}{e^x+1} \\ &= -\frac{1-e^x}{1+e^x} \\ &= -f(x) \end{aligned}$$

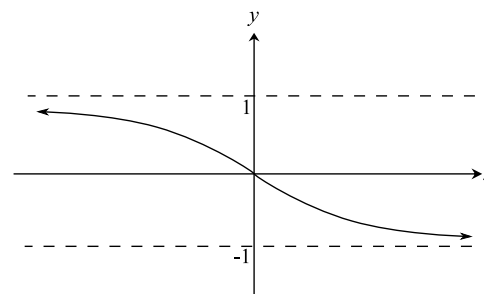
$\therefore f(x)$  is ODD.

ii.  $y = \frac{1-e^x}{1+e^x} \times \frac{e^{-x}}{e^{-x}} = \frac{e^{-x}-1}{e^{-x}+1} = 1 - \frac{2}{e^{-x}+1}$

As  $x \rightarrow \infty$ ,  $\frac{2}{e^{-x}+1} \rightarrow 2$ ,  $y \rightarrow -1$

As  $x \rightarrow -\infty$ ,  $\frac{2}{e^{-x}+1} \rightarrow 0$ ,  $y \rightarrow 1$

When  $x = 0$ ,  $y = 0$



## 11. L&E, 2ADV E1 2024 HSC 13

|                                            | Population $W$ | Population $K$ |
|--------------------------------------------|----------------|----------------|
| Population in 1985                         | $A = 34$       | $B = 280$      |
| Percentage yearly change in the population | $+5.5\%$       | $-3.0\%$       |
| Predicted population when $x = 50$         | 494            | 61             |

Calculations:

Population  $W$ :  $(1.055)^x \Rightarrow$  increase of  $5.5\%$  each year

Population  $K$ :  $(0.97)^x \Rightarrow 1 - 0.97 = 0.03 \Rightarrow$  decrease of  $3.0\%$  each year

Predicted population ( $W$ ) =  $34 \times (1.055)^{50} = 494$

## 12. Algebra, STD2 A4 2004 HSC 26a

i. Percentage increase

$$= \frac{114 - 100}{100} \times 100$$

$$= 14\%$$

**COMMENT:** Common ADV/STD2 content in new syllabus.

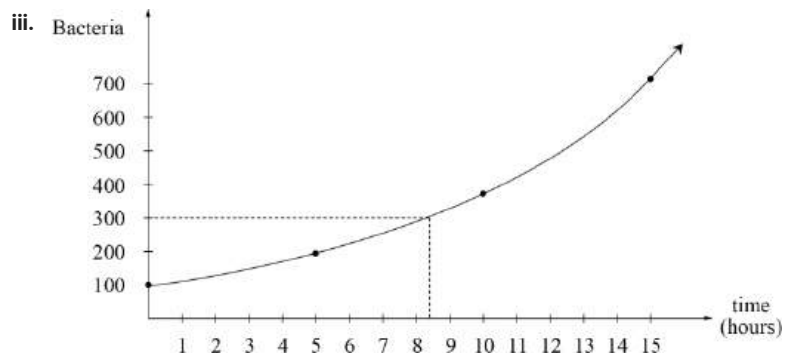
ii.  $n = 100(1.14)^t$

When  $t = 15$ ,

$$n = 100(1.14)^{15}$$

$$= 713.793 \dots$$

$$= 714 \text{ (nearest whole)}$$



iv. Using the graph

The number of bacteria reaches 300 after approximately 8.4 hours.

## 13. Calculus, 2ADV C3 2019 MET1 4

a. Domain:  $x > 3$

Range:  $y \in \mathbb{R}$

b.i.  $g(x) = \log_e(x - 3) + 2$

$$g'(x) = \frac{1}{x - 3}$$

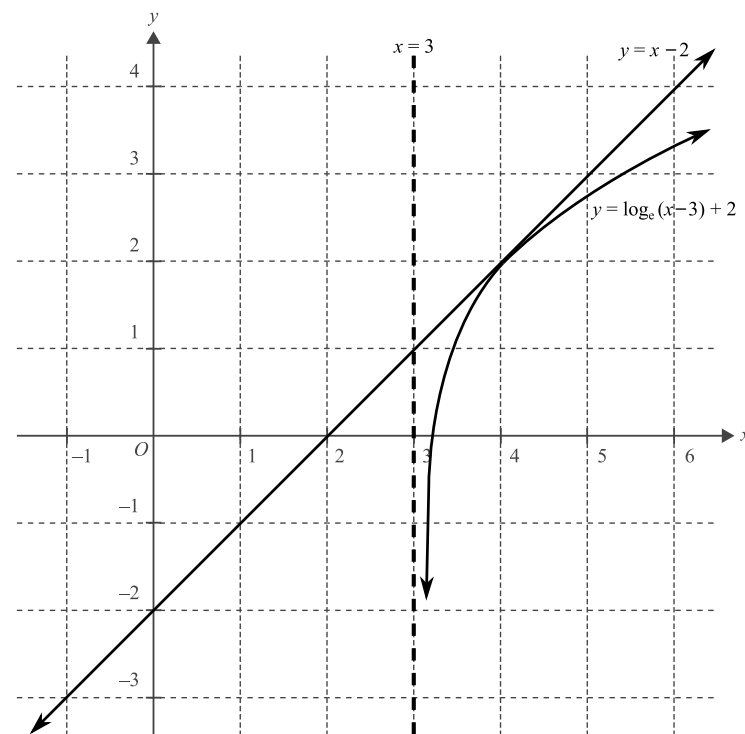
$$g'(4) = 1$$

Equation of tangent,  $m = 1$  through  $(4, 2)$ :

$$y - 2 = 1(x - 4)$$

$$y = x - 2$$

b.ii.

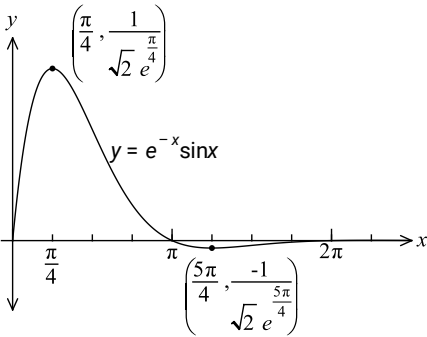


14. Calculus, 2ADV C3 2023 HSC 30

**a.**  $f(x) = e^{-x}\sin x$   
 $f'(x) = e^{-x}\cos x - e^{-x}\sin x = e^{-x}(\cos x - \sin x)$   
SPs when  $f'(x) = 0$ :  
 $e^{-x} = 0 \rightarrow$  no solution  
 $\cos x - \sin x = 0$   
 $1 - \tan x = 0$   
 $\tan = 1$   
  
 $x = \frac{\pi}{4}, \frac{5\pi}{4}$   
  
 $f(\frac{\pi}{4}) = e^{-\frac{\pi}{4}}\sin \frac{\pi}{4} = \frac{1}{\sqrt{2} \times e^{\frac{\pi}{4}}}$   
 $f(\frac{5\pi}{4}) = e^{-\frac{5\pi}{4}}\sin \frac{5\pi}{4} = \frac{-1}{\sqrt{2} \times e^{\frac{5\pi}{4}}}$   
  
 $\therefore$  SPs at  $(\frac{\pi}{4}, \frac{1}{\sqrt{2} \times e^{\frac{\pi}{4}}})$  and  $(\frac{5\pi}{4}, \frac{-1}{\sqrt{2} \times e^{\frac{5\pi}{4}}})$

Mean mark (a) 54%.

**b.**  $x$ -intercepts at  $x = 0, \pi, 2\pi$



♦♦ Mean mark (b) 34%.

15. Algebra, STD2 A4 2012 HSC 30c

- i.** 2030 occurs at  $n = 20$  on the  $x$ -axis.  
Expected population (no policy) = 6 600 000
- ii.**  $A$  represents the population when  $n = 0$   
which is the population in 2010.
- iii. (1)**  $P = A(1.05)^n$  would be steeper and lie above  
 $P = A(1.04)^n$  since  $1.05 > 1.04$

**COMMENT:** Common ADV/STD2 content in new syllabus.

♦ Mean mark part (ii) 48%

- iii. (2)** Let  $b = 1.03$   
 $P = 3\,000\,000 \times 1.03^{20}$   
 $= 5\,418\,000$   
  
Let  $b = 1.02$   
 $P = 3\,000\,000 \times 1.02^{20}$   
 $= 4\,457\,800$

$\therefore b = 1.02$

- iv.** In 2050,  $n = 40$   
 $P = 3\,000\,000 \times 1.02^{40}$   
 $= 6\,624\,119$  (nearest whole)

Since the population is below 7 million,  
the model will achieve the aim.

16. Algebra, STD2 A4 2016 HSC 29b

- i.** When  $x = 0$ ,  
 $1.5 = A(1.1)^0$   
 $\therefore A = 1.5$  kg
- ii.** Daily growth rate  
 $= 0.1$   
 $= 10\%$

♦ Mean mark (i) 48%.  
♦♦♦ Mean mark part (ii) 6%!